Conic Relaxations in Distribution Systems

Hybrid Exactness via Graph Decomposition

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2026 INFORMS Annual Meeting
Nov. 1, 2026

Overview

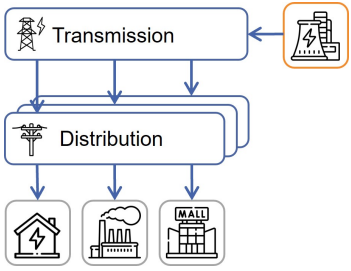


Figure 1: Traditional Energy Systems

Overview

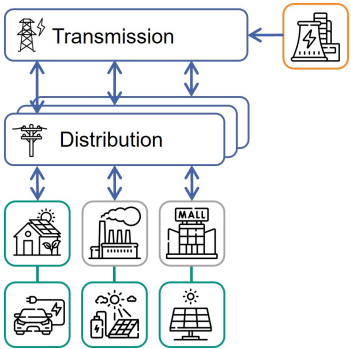


Figure 1: Modern Energy Systems

- Modern Energy Systems
  - Distributed Energy Resources
  - Bidirectional Power Flow
  - Tractable Real-Time Computation
- Conic Relaxation in Distributions
  - Fast Computation
  - Lower bound for objective
  - Globality if exact

DistFlow Equations for Tree Graph

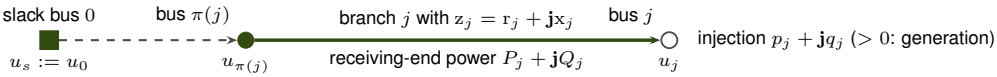


Figure 2: Illustration of the DistFlow model.

- Tree rooted at 0, with branches oriented away from 0.
- Branch j start from its parent $\pi(j)$ and ends at bus $j \in \mathcal{N} = \{1, \cdots, n\}$.
- Squared magnitude of branch current and nodal voltage

$$\ell_j := |I_j|^2 \qquad \text{and} \qquad u_j := |V_j|^2$$

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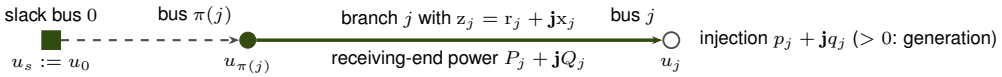


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DistFlow Equations [1], [2]

$$p_j = \sum_{k:\pi(k)=j} (P_k + \mathrm{r}_k \ell_k) - P_j \qquad \forall j \in \mathcal{N} = \{1, \cdots, n\} \qquad \text{(1a)}$$

$$q_j = \sum_{k:\pi(k)=j} (Q_k + \mathrm{x}_k \ell_k) - Q_j \qquad \forall j \in \mathcal{N} = \{1, \cdots, n\} \qquad \text{(1b)}$$

$$u_j = u_{\pi(j)} - 2(\mathrm{r}_j P_j + \mathrm{x}_j Q_j) - |\mathrm{z}_j|^2 \ell_j \qquad \forall j \in \mathcal{N} = \{1, \cdots, n\} \qquad \text{(1c)}$$

$$u_j \ell_j = P_j^2 + Q_j^2 \qquad \forall j \in \mathcal{N} = \{1, \cdots, n\} \qquad \text{(1d)}$$

[1] M. Baran and F. F. Wu, "Optimal capacitor placement on radial distribution systems," *IEEE Transactions on Power Delivery*, 1989

[2] M. Baran and F. F. Wu, "Optimal sizing of capacitors placed on a radial distribution system," *IEEE Transactions on Power Delivery*, 1989

Convex Relaxation of Optimal Power Flow

Original Problem

$$\min_{z \in \mathcal{Z}} f(z) \tag{2a}$$

$$\text{s.t. } g_{\text{L}}(z) = 0 \quad | \lambda_{\text{L}}, \tag{2b}$$

$$g_{\text{N}}(z) = 0 \quad | \lambda_{\text{N}}. \tag{2c}$$

- $z = (P, Q, u, \ell, p, q) \in \mathbb{R}^{6n}$: stacked decision variables;
- $\mathcal{Z} = \{z \,|\, z_{\text{MIN}} \leq z \leq z_{\text{MAX}}\}$: convex set of operating limits;

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- $\mathcal{Z} = \{z \,|\, z_{\text{MIN}} \leq z \leq z_{\text{MAX}}\}$: convex set of operating limits;
- The linear functions $g_{\text{L}} : \mathbb{R}^{6n} \to \mathbb{R}^{3n}$ represents the affine constraints (1a)–(1c), with corresponding Lagrange multipliers $\lambda_{\text{L}} \in \mathbb{R}^{3n}$;
- The nonlinear functions $g_{\text{N}} : \mathbb{R}^{6n} \to \mathbb{R}^{n}$ represents the quadratic constraints (1d), with corresponding Lagrange multipliers $\lambda_{\text{N}} \in \mathbb{R}^{n}$;

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- Constraint (4c) represents second-order cone constraints.

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Definition 2.4 (Exactness [3, Def. 1])

The relaxed problem (4) is *exact* if *every* optimal solution z^* satisfies $g_{\mathrm{N}}(z^*) = 0$.

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- Exactness $\Rightarrow$ a convex solver returns a global optimizer of the nonconvex OPF.
- Exactness is not automatic; it requires conditions on the controls, network, limits, and objective.

Two Classical Exactness Certificates

I. *Dispatchable* systems [4]

- Requires, among other assumptions, that **all** nodal injections (p, q) be **controllable**.
- Perturbation: *hold all voltages fixed* and rebalance power injections at the incident buses.
- The effect remains localized to that branch.

II. *Passive* systems [3]

- Requires a *network-wide* **path condition** C1 [3, Thm. 1].
- Allows power injections to be **fixed**.
- Perturbation: hold injections fixed, allowing *voltages to change throughout the network*.

[4] M. Farivar and S. H. Low, "Branch flow model: Relaxations and convexification—Part I," *IEEE Transactions on Power Systems*, vol. 28, no. 3, pp. 2554–2564, 2013
[3] L. Gan, N. Li, U. Topcu, and S. H. Low, "Exact convex relaxation of optimal power flow in radial networks," *IEEE Transactions on Automatic Control*, 2015

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Sufficient conditions for C1 [3, Thm. 4]

- No reverse power flows, i.e.,

$$P_j,\ Q_j \geq 0, \qquad \forall j \in \mathcal{N}.$$

- Constant, nondecreasing, or nonincreasing $\mathrm{r}_j/\mathrm{x}_j$ ratios from the slack bus toward the leaves.

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Same mechanism, different footprint

- Both analyses start from a relaxed solution (4) and construct a perturbation that lowers the cost.
- They differ in what is held fixed: *voltages* in (I) or *injections* in (II).
- In (II), voltage changes can propagate to downstream buses.

[4] M. Farivar and S. H. Low, "Branch flow model: Relaxations and convexification—Part I," *IEEE Transactions on Power Systems*, vol. 28, no. 3, pp. 2554–2564, 2013
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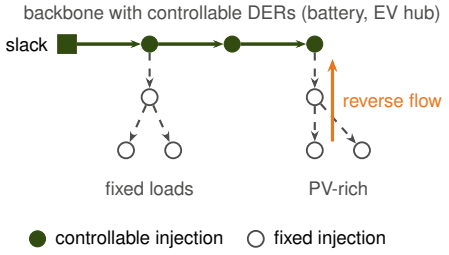
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- Limitations of *network-wide* certificates:

Figure 3: A partially dispatchable feeder.

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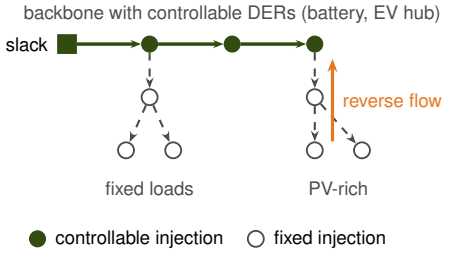


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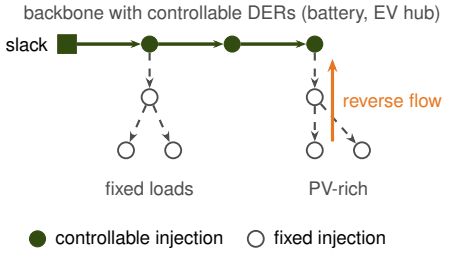


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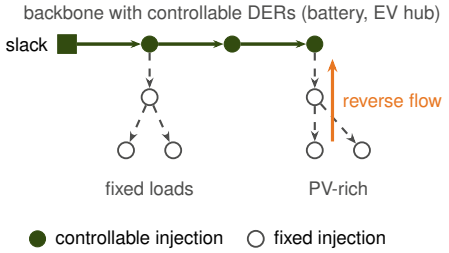


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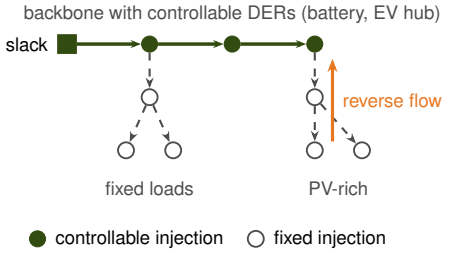


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  - Certify each region, then combine the results.

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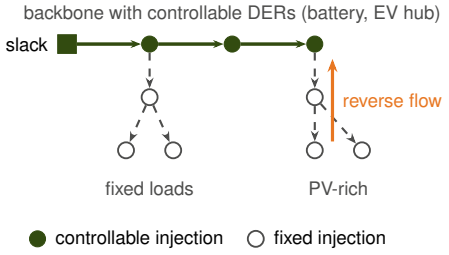


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  - Certify each region, then combine the results.
  - Challenge: a perturbation in one passive region may change the **root voltages** of downstream regions.

Ingredient 1: Strictly Positive Multipliers

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$$\min_{z \in \mathcal{Z}} f(z) \tag{3a}$$

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- With $w := (z, \lambda_{\mathrm{L}}, \lambda_{\mathrm{N}})$ and KKT mapping $\mathscr{F}(w) := \big(\nabla_z \mathcal{L},\ -g_{\mathrm{L}},\ -g_{\mathrm{N}}\big)$, the two KKT systems differ **only in the cone of λ_{N}**:

$$0 \in \mathscr{F}(w) + N_{\mathcal{Z}\times\mathbb{R}^{3n}\times\mathbb{R}^{n}}(w) \ \text{ (original)}, \qquad 0 \in \mathscr{F}(w) + N_{\mathcal{Z}\times\mathbb{R}^{3n}\times\mathbb{R}^{n}_{+}}(w) \ \text{ (relaxed)}.$$

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Theorem 4.1 (Exactness via nonlinear multipliers)

Let $w^* = (z^*, \lambda_{\mathrm{L}}^*, \lambda_{\mathrm{N}}^*)$ be a KKT point of the OPF (3) with $\lambda_{\mathrm{N}}^* \in \mathbb{R}^n_{++}$. Then w^* is a KKT point of the relaxation (4), z^* is a global minimizer of both, and with fixed slack voltage, z^* is unique.

[5] L. Gan, N. Li, U. Topcu, and S. Low, "On the exactness of convex relaxation for optimal power flow in tree networks," *51st IEEE Conference on Decision and Control (CDC)*, 2012

Ingredient 2: Lagrangian Multiplier

At a KKT point, select the **normal vector** $\xi^* := \nabla f(z^*) + \mathrm{D}g_{\mathrm{L}}(z^*)^\top \lambda_{\mathrm{L}}^* + \mathrm{D}g_{\mathrm{N}}(z^*)^\top \lambda_{\mathrm{N}}^* \in N_{\mathcal{Z}}(z^*)$.

Proposition 4.2 (Individual multiplier formula)

At a KKT point w^*, suppose δ^j satisfies $\mathrm{D}g_{\mathrm{L}}(z^*)\delta^j = 0,\ \mathrm{D}g_{\mathrm{N}}(z^*)\delta^j = \eta \mathbf{e}_j$, with $\eta \neq 0$. Then

$$[\lambda_{\mathrm{N}}^*]_j = -\frac{\langle \nabla f(z^*) + \xi^*, \delta^j \rangle}{\eta}. \tag{5}$$

If $\langle \nabla f(z^*), \delta^j \rangle < 0,\ \delta^j \in T_{\mathcal{Z}}(z^*),\ \eta > 0$, then

$$[\lambda_{\mathrm{N}}^*]_j > -\frac{\langle \nabla f(z^*), \delta^j \rangle}{\eta} > 0. \tag{6}$$

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- Generalized-equation version of the classical sensitivity interpretation of Lagrange multipliers [6, Prop. 3.2.2] [7, Cor. 4.8]: δ^j is a **perturbation direction** that isolates the sign of one multiplier $[\lambda_{\mathrm{N}}^*]_j$.
- Exactness becomes a **branchwise** property that can be certified **region by region**.

[6] D. Bertsekas, *Nonlinear Programming*, 2nd ed. Athena Scientific, 1999

[7] F. Pacaud, "Sensitivity analysis for parametric nonlinear programming: A tutorial," *arXiv preprint arXiv:2504.15851*, 2025

Ingredient 3: Regularity and Nonnegative Inverse

- The partial Jacobian of the DistFlow constraints g with respect to (P, Q, u, ℓ) is

$$J(z) := \partial_{(P,Q,u,\ell)} g(z; u_s) \in \mathbb{R}^{4n \times 4n}.$$

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$$\det(J(z)) \neq 0.$$

- We require regularity for the network and every connected subtree:
$$\det(J_\kappa(z_\kappa)) \neq 0 \qquad \forall \kappa \in \mathcal{S}(\mathcal{G}),$$
with z_κ, J_κ the local variables and Jacobian of subtree κ, and $\mathcal{S}(\mathcal{G})$ the set of all connected subtrees of $\mathcal{G}$.

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- Subtree Regularity:
$$\det(J_\kappa(z_\kappa)) \neq 0 \qquad \forall \kappa \in \mathcal{S}(\mathcal{G}).$$

- Not restrictive: implied by standard sufficient conditions for power flow regularity [8, Thm. 1],[9, Thm. 10], (see [10, Prop. 3.5, Cor. 3.12]).

[8] Z. Wang, B. Cui, and J. Wang, "A necessary condition for power flow insolvability in power distribution systems with distributed generators," *IEEE Transactions on Power Systems*, 2017

[9] L. Aolaritei, S. Bolognani, and F. Dörfler, "A distributed voltage stability margin for power distribution networks,", 2017, 20th IFAC World Congress

[10] X. Dai, "Distflow and its conic relaxation: Regularity, monotonicity, and hybrid exactness via graph decomposition,",

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- **Nonnegative inverse:** if $z \in \mathcal{R} \cap \mathcal{C}$ (regular set ∩ path-condition set), the inverse of the condensed sensitivity matrix is nonnegative [10, Props. 3.10, 3.19], which provides the **signs** of the perturbation directions.

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Partition into Dispatchable and Passive Subtrees

Partition rule (Section 5.1)

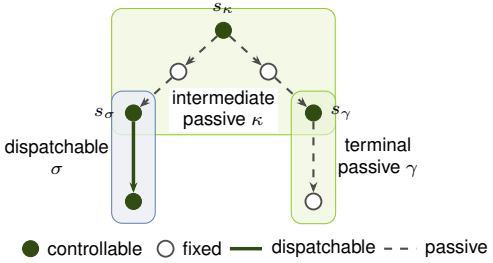


Figure 4: Graph decomposition: the children σ, γ are rooted at buses in subtree κ, as illustrated in [10, Fig. 1(a)].

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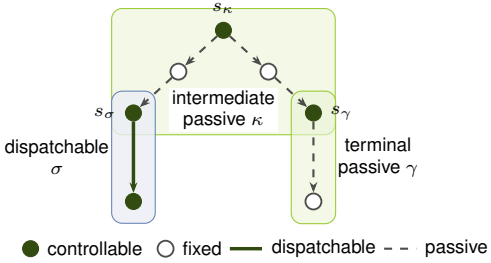


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(iii) **Terminal** passive subtree $\mathcal{K}_{\mathrm{tp}}$: nothing is attached downstream

(iv) **Intermediate** passive subtree $\mathcal{K}_{\mathrm{tp}}$: at least one bus attached downstream

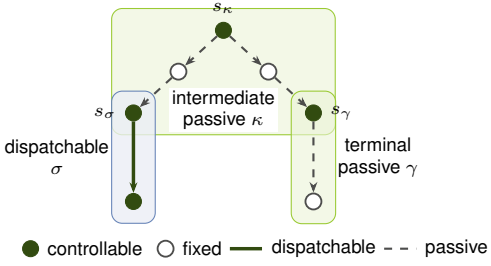


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Hybrid exactness via Graph Decomposition



Figure 5: Perturbation direction in a dispatchable system.

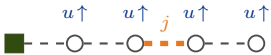


Figure 6: Perturbation direction in a passive system.

Hybrid exactness via Graph Decomposition



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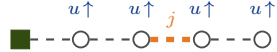


Figure 6: Perturbation direction in a passive system.

Lemma 5.1 (Responses to a root-voltage change $\tau := \delta_{u_s} > 0$)

Type	Fixed	Condition	Voltages $u \in \mathbb{R}^n$	Current $\ell \in \mathbb{R}^n$	Total-loss $\mathrm{x}^\top \ell \in \mathbb{R}$	Slack injection $p_0 \in \mathbb{R}$
Passive	(p, q)	$z \in \mathcal{R} \cap \mathcal{C}$	$\delta_u - \tau \in \mathbb{R}^n_+$	–	$\mathrm{r}^\top \delta_\ell \in \mathbb{R}_+$	$\delta_{p_0} \in \mathbb{R}_-$
Dispatchable	(P, Q)	$z \in \mathcal{R}$	$\delta_u - \tau \in \mathbb{R}^n_+$	$\delta_\ell \in \mathbb{R}^n_+$	$\mathrm{r}^\top \delta_\ell \in \mathbb{R}_+$	$\delta_{p_0} \in \mathbb{R}_-$

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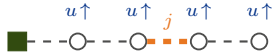


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Theorem 5.3 (Hybrid exactness)

For a KKT point $(z^*, \lambda^*_{\mathrm{L}}, \lambda^*_{\mathrm{N}})$ of the OPF (3) with $z^* \in \mathcal{R}$, $\mathrm{r} \in \mathbb{R}^n_{++}$, partition $\mathcal{K}$. Assume

- **local:** $z^*_\kappa \in \mathcal{C}_\kappa$ for every passive subtree $\kappa \in \mathcal{K}_{\mathrm{p}}$;
- **tangent:** $\delta^j \in T_{\mathcal{Z}}(z^*)$ for every branch;
- **objective:** either (i) $\nabla_{(p;p_0)} f(z^*) \in \mathbb{R}^{n+1}_{++}$, or (ii) $\nabla_\ell f(z^*) = \alpha\, \mathrm{r} \in \mathbb{R}^n_{++}$ with some $\alpha > 0$ and $\nabla_u f(z^*) \in \mathbb{R}^n_-$.

Then $\lambda^*_{\mathrm{N}} \in \mathbb{R}^n_{++}$, and the relex problem (4) is exact. z^* is a global minimizer of both problems (3)(4), unique when u_s is fixed.

Where to Install Controllable DERs

Reverse flow is harmless on a dispatchable branch

Controllable: injections at *both* ends make a branch dispatchable;
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Figure 7: Reverse flow on a passive branch.

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Figure 7: Reverse flow on a passive branch.



Figure 8: Reverse flow on a dispatchable branch.

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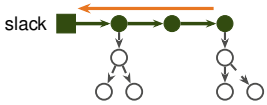


Figure 9: A partially dispatchable feeder.

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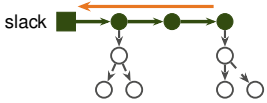


Figure 9: A partially dispatchable feeder.

Corollary 5.4 (Dispatchable backbone with terminal passive subtrees)

Dispatchable backbone connected to the slack bus, all passive subtrees terminal $\Rightarrow$ Theorem 5.3 holds, and a passive direction changes branch variables only inside κ and the injection only at s_κ: $\mathcal{F}_\kappa$ / $\mathcal{C}_\kappa$ are checked *subtree by subtree*. (Caveat: dispatchability removes the sign tests on a branch, not the tangency and objective conditions.)

Conclusion & Outlook

Conclusion

Outlook

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[11] H. Moring, D. K. Molzahn, and J. L. Mathieu, "Exploring the inexactness of second-order cone relaxations for calculating operating envelopes," *IEEE Transactions on Power Systems*, 2026

Conclusion & Outlook

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- **Branchwise exactness:** allow for analysis region by region.
- **Composition:** dispatchable/passive partition, hybrid certificate and controllable DERs at both ends for a reverse flow branch.

Outlook

- Degeneracy analysis: when power flow regularity is not satisfying or when critical operating limit are binding
- Meshed grids: tree path structure lost; cycle constraints enter.
- Failed regional test: can the decomposition localize and bound the inexactness? [11]
- Unbalanced networks, shunts, tap changers.

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